

Postulates, Theorems, Converses, and Biconditionals

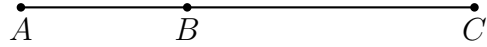
postulates and theorems, converses, and biconditionals

- A **postulate** is accepted without proof; a **theorem** is proved from postulates, definitions, and earlier theorems.
 - The **converse** of “if P then Q ” is “if Q then P ”. A converse may be true or false, and it has to be judged on its own.
 - A **biconditional** (“ P if and only if Q ”) asserts both directions at once, so it is only available when the statement *and* its converse are true.
 - Every diagram shows labelling only; all measurements are in the problem text.
-

- 1.** Label each statement **P** (postulate) or **T** (theorem), then write one sentence naming the test you used.

1	Through any two points there is exactly one line.	
2	Vertical angles are congruent.	
3	If two points lie in a plane, the line through them lies in that plane.	
4	If two angles are supplements of the same angle, they are congruent.	

2. B lies between A and C , so the Segment Addition Postulate gives $AB + BC = AC$. In centimetres, $AB = 2x + 3$, $BC = x - 1$, and $AC = 20$. Find x .



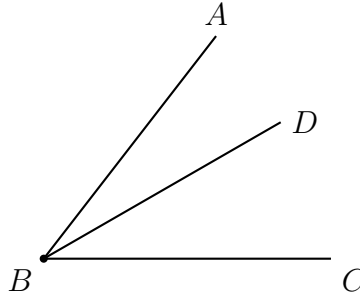
Answer: _____

3. Consider the theorem: *“If two angles are vertical angles, then they are congruent.”*

(a) Write the converse of this theorem.

(b) Is the converse true? If it is, say why; if it is not, give a counterexample.

4. $\overrightarrow{BD}$ lies in the interior of $\angle ABC$, so the Angle Addition Postulate gives $m\angle ABD + m\angle DBC = m\angle ABC$. In degrees, $m\angle ABD = 3x+5$, $m\angle DBC = 2x-5$, and $m\angle ABC = 65$.



- (a) Find x .

Answer: _____

- (b) Find $m\angle DBC$, the smaller of the two parts.

Answer: _____ degrees

5. A textbook states: “An angle is a right angle if and only if its measure is 90° .”

- (a) Write the two conditional statements this biconditional combines.

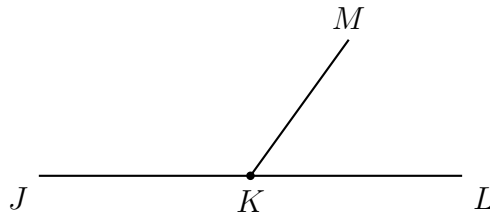
- (b) Explain why a biconditional is available here but would not be available if only one of those two conditionals were true.

6. Consider the postulate: “If two angles form a linear pair, then they are supplementary.”

(a) Write the converse.

(b) Decide whether the converse is true, and justify your decision.

7. Two angles form a linear pair, so by the Linear Pair Postulate they are supplementary. In degrees the two angles measure $5x - 8$ and $3x + 12$. Find x .



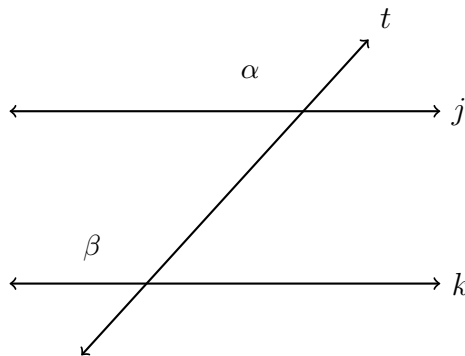
Answer: _____

8. The definition of congruent segments says: “If two segments are congruent, then they have the same length.” Its converse is also true.

(a) Write the biconditional the two directions allow, and say what has to hold before you are entitled to write it.

(b) Choose one statement from earlier on this worksheet that may *not* be rewritten as a biconditional, and justify your choice.

9. The Converse of the Corresponding Angles Postulate says: if corresponding angles are congruent, then the lines are parallel. In the figure, the two marked corresponding angles measure $(4x + 15)^\circ$ and $(6x - 25)^\circ$. Find the value of x for which $j \parallel k$.



Answer: _____

10. “Two lines are perpendicular if and only if they form a right angle.”

(a) Write the two conditionals this combines, and explain which direction you use when perpendicular lines are *given* and which you use when perpendicular lines must be *concluded*.

Answer: _____

(b) Two lines meet forming an angle of $(7x + 6)^\circ$. Find the value of x that entitles you to conclude the lines are perpendicular.

Answer: _____